

Zebrafish DEA March 2026

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Sample Preparation and RNA Extraction/Sequencing (NRC)

- RNA was isolated from zebrafish embryos that had been treated with potential neuroprotective agents in a BMAA chemical model of neurodegeneration
- Libraries were prepared using the Illumina Truseq Stranded RNA kit which uses PolyA enrichment to remove rRNA
- Sequencing was performed on an Illumina NextSeq 2000 platform
- Performed in two rounds of 3 replicates for each of the 5 conditions
- Total of 30 biological samples (6 reps x 5 conds)
- Each sample underwent paired-end sequencing and generated two files: R1 (forward read) and R2 (backward read) for a total of 60 FASTQ files

Count Generation (Gurkamal)

These files were received from gurkamal@canurta.com on January 5th 2026 in a email subject: "Drosophila so far". The files attached are all pertaining to the zebrafish study. The explanation of how files were generated comes from a MS Teams message sent on January 9th 2025, archived at [Kamal Jan 9th 2026](#). "For the zebrafish dataset, I worked with an R environment in the cloud so it was long formatted for R."

Step	Tool	Input	Output	Notes
QC	FastQC + MultiQC	Raw paired-end reads (R1 and R2 aggregated into 30 FASTQ files)	Assessments of read quality, adapter content, duplication, and GC distributions	
Trimming	fastp	Raw reads (30 FASTQ files)	Trimmed reads (30 FASTQ files)	Trimmed low-quality bases with Phred < 20 (Phred is a measure of how likely a base call is wrong; Phred < 20 means 1% error rate) and reads with <36 base pairs (common minimum for vertebrate genomes; less than this is not alignable)
Alignment	HISAT2	Raw reads (30 FASTQ files) Reference genome	30 SAM/BAM files	Downloaded the GRCz11 reference genome and the complementary ensemble GTF (annotation). - Used HISAT2 (splice-aware) index to produce SAM (sequence alignment/map) files and then samtools to convert SAM to BAM (binary alignment/map) files for machine readability.
Quantification	featureCounts	30 BAM files	raw_counts.txt raw_counts.txt.summary	Reads that mapped ambiguously or with low confidence were automatically ignored.
Parsing	Pandas	raw_counts.txt	counts_deseq2.txt (genes x samples)	Removed the extra columns from the featureCounts table and renamed the sample columns from BAM file paths to sample IDs so the count matrix could be used in DESeq2 (genes x samples). This needs to be transposed (samples x genes) for pydeseq2.

Sample-Level Quality Control

Sample Screening

These 4 samples were flagged for violating relative thresholds (beyond 2 MAD from the median in incorrect direction). No absolute thresholds were violated.

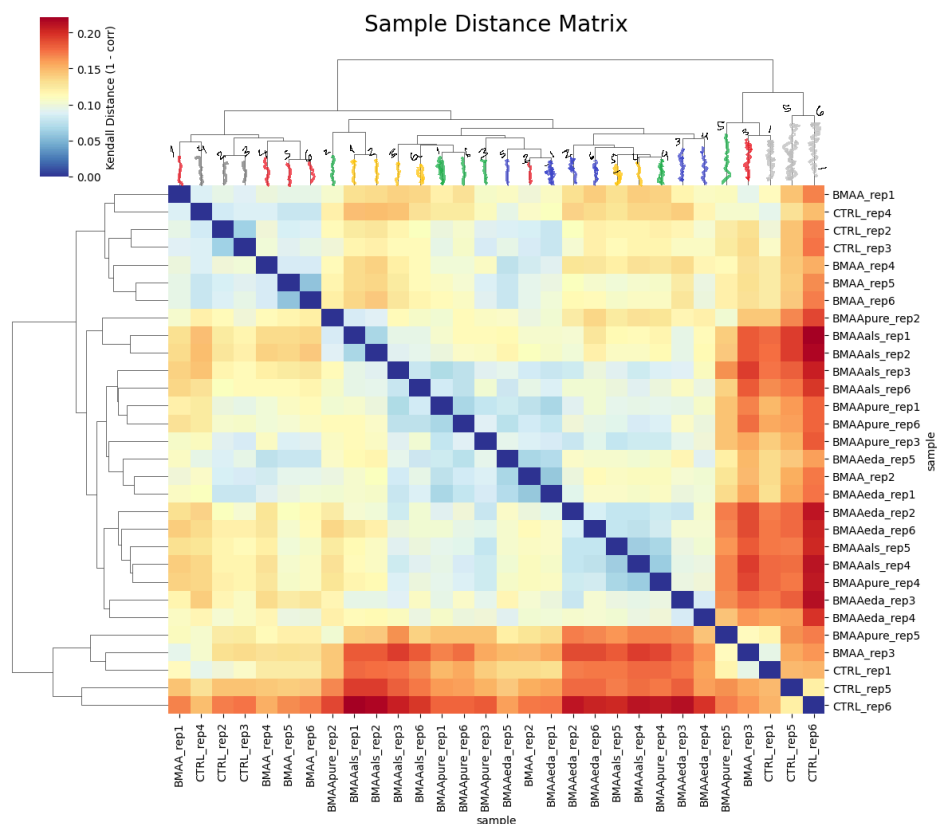
sample	relative_thresholds	total_reads	library_size	mapping_rate	n_detected_gei
BMAA_rep3	low_mapping_rate	35654113	23773774	0.6667891022839356	25449
BMAAeda_rep6	low_n_detected_genes	25700188	18536313	0.721252039090142	24226
BMAApure_rep2	low_mapping_rate,high_multimapping_rate	30028598	20369751	0.6783450562693603	25421
CTRL_rep2	high_multimapping_rate	39329094	27539680	0.7002368272200727	25242

Sample Correlation Heatmap

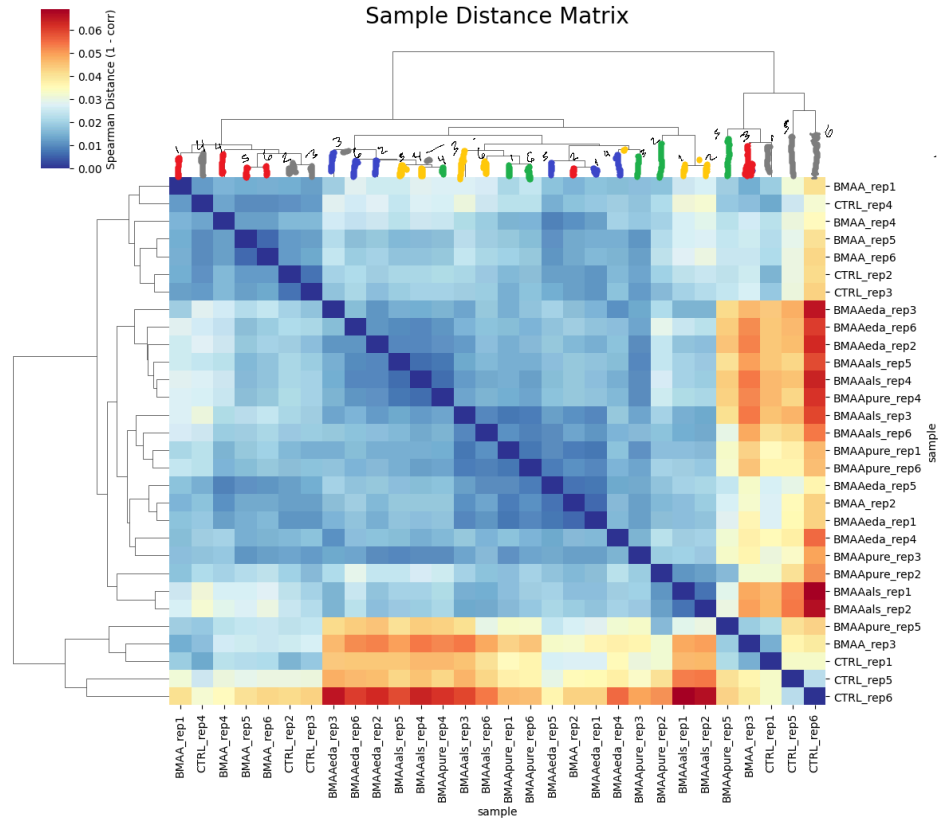
A sample correlation heatmap of all samples shows poor clustering, with controls (healthy and pathogenic) grouping together and all treated samples in another large group. The left control cluster is mostly pathogenic (drug=BMAA) and the right is mostly healthy (drug=none). Based on this assumption, the following outliers emerge:

- CTRL_rep4, CTRL_rep2, and CTRL_rep3 group with pathogenic controls
- BMAA_rep3 groups with healthy controls
- BMAA_rep2 is in the treated group
- BMAApure_rep5 groups with healthy controls

After PCA, the healthy controls 5 and 6 on the right are batch 3, making the other 3 samples in this cluster outliers. Removing these 3 leaves 2 clusters: control (healthy and pathogenic) and treated. Within controls, the lack of distinction between pathogenic and control is probably sex (probably PC1). Under this assumption, the pathogenic vs healthy distinction does appear under the parent sex bifurcation.



Similar patterns are achieved with a spearman method. Outliers remain the same. More distance is shown between the control group on the right (more red, first branch) indicating this is a weaker group.

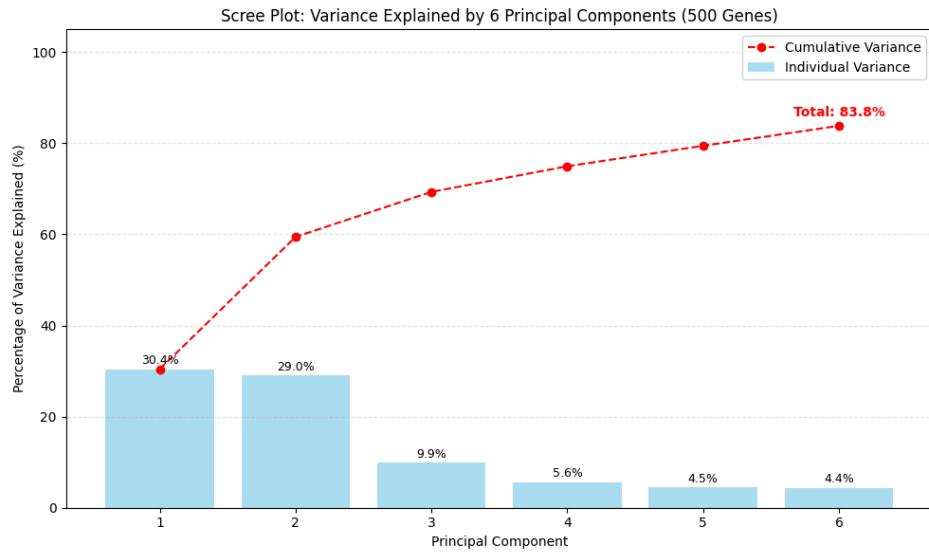


Principal Component Analysis

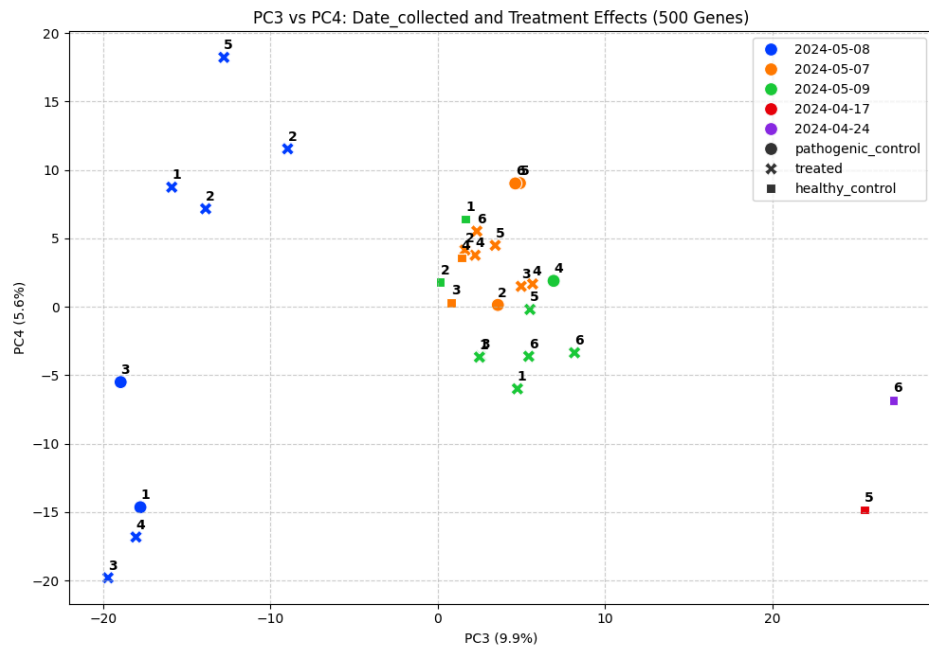
PC	Variance Explained	Experimental Variable
1	30.4%	Probably sex
2	29.0%	?
3	9.9%	Batch effect (date collected)

Variation is mostly explained by PC1 and PC2.

Experiment (reps 1-3 vs reps 4-6) doesn't seem to appear on the first 4 PCs, so it doesn't need to be accounted for in design formula.

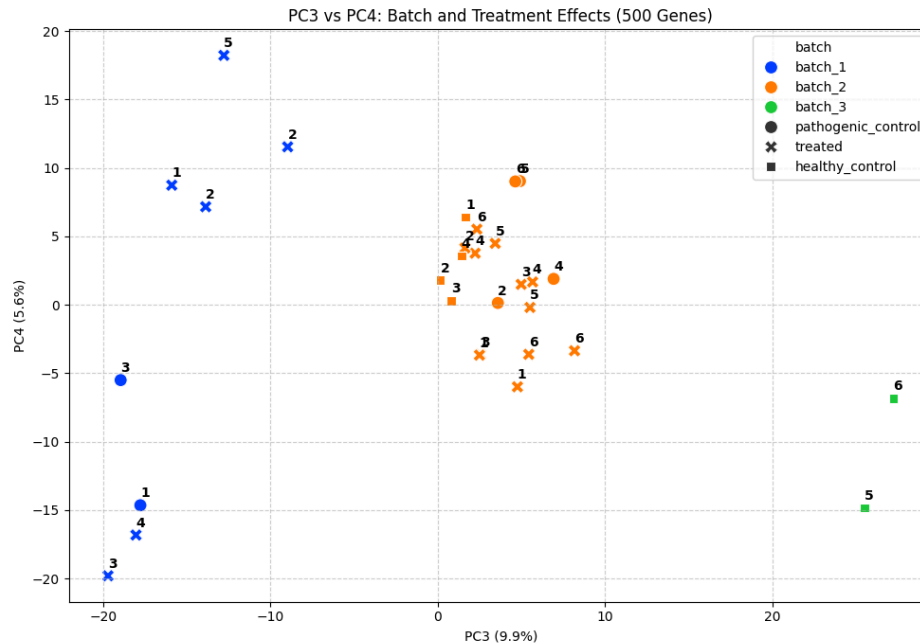


PC3 looks like batch effects from date of specimen collection. Note that batch 3 containing CTRL_rep5 and CTRL_rep6 are the most different samples on the heatmap (far right). This exposes BMAApure_rep5, BMAA_rep3, and CTRL_rep1 as potential outliers. Consider removing these 3, since an experimental variable can't be found (and thus statistically controlled for) for these samples.



- Batch 1: May 8th
- Batch 2: May 7th and May 9th
- Batch 3: April 17th and April 24th

Added to the metadata to be controlled for in the design formula.

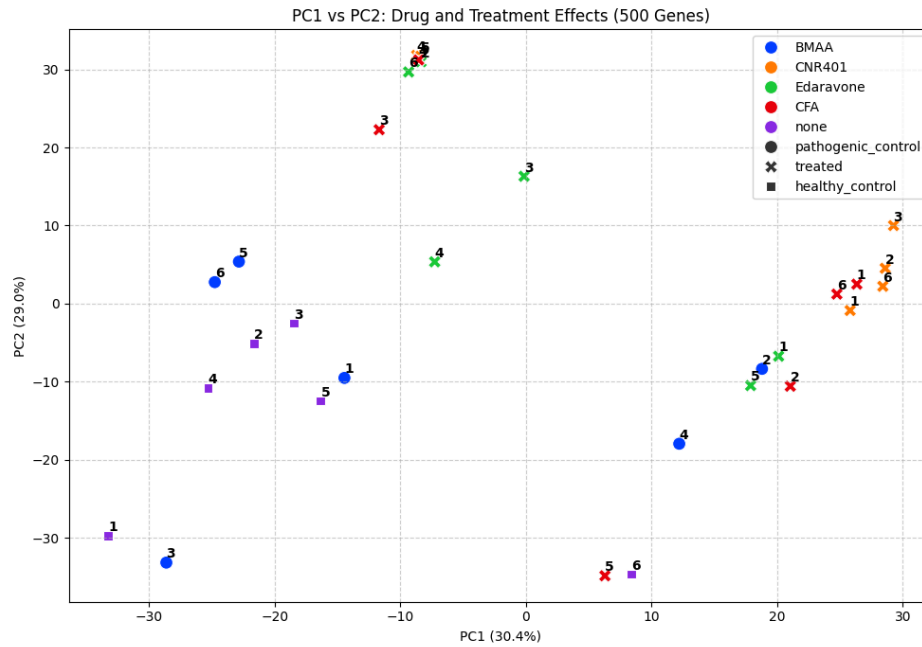


PC1 doesn't look to separate by drug or treatment (vs controls). This could very well be sex which was not recorded, evidenced by the drosophila dataset. An inferred sex column could be added to the metadata to account for this in the design formula. However, the two groups under the controls do not separate by PC1 (inferred sex): BMAA_rep1 and CTRL_rep4 (one strata) are near CTRL_rep2, CTRL_rep3, BMAA_rep5, and BMAA_rep6 (other strata). Based on this, I don't think I should apply an artificial sex variable to the metadata.

BMAA and healthy samples could be clustering together because BMAA (a neurotoxin) acts on the protein level and doesn't cause a transcriptomic impact. This theory is supported by the fact that locomotion was observed to be impaired. With the phenotypic proof, the transcriptomic similarity of healthy and diseased groups isn't as worrying.

PC2 could be treated (vs controls) where drug-treated is higher on PC2. Under this assumption:

- BMAApure_rep5 groups with controls



Sample	Relative (MAD) violation	Correlation violation	PCA notes
BMAA_rep3	low_mapping_rate	Once batch 3 is accounted for, exposed as an outlier.	
BMAAeda_rep6	low_n_detected_genes		
BMAApure_rep2	low_mapping_rate,high_multimapping_rate		
CTRL_rep2	high_multimapping_rate	clusters with left controls	
CTRL_rep3		clusters with left controls	
CTRL_rep4		clusters with left controls	
BMAApure_rep5		Once batch 3 is accounted for, exposed as an outlier.	clusters with controls on PC2
BMAA_rep2		clusters with treated groups	
CTRL_rep5		Largest distance	Batch 3
CTRL_rep6		Largest distance	Batch 3
CTRL_rep1		Once batch 3 is accounted for, exposed as an outlier.	

Sample QC Conclusion

- CTRL_rep1, BMAA_rep3 (also low mapping rate), BMAApure_rep5 (also PC2 outlier) will be removed from DEA.
- Batch effects will be adjusted for in the design formula
- Sex cannot be accounted for, which is a limitation of this study. However, differentially expressed genes/gene sets that are caused by sex should be easy to manually pick out and exclude from discussion. To preserve power and avoid mislabeling samples, an artificial sex label will not be applied.

Differential Expression Analysis

Gene and Sample Filtering

There were initially 27 samples after dropping the 3 outliers above, one of which is a healthy control. Then, healthy controls were dropped, eliminating batch 3 and leaving 22 samples. After gene filtering (<10 counts), 26209 genes were retained (32520 - 6311).

Metadata dataframe that entered the `DeseqDataSet`.

sample	drug	treatment	replicate	batch
BMAA_rep1	BMAA	pathogenic_control	1	batch_1
BMAA_rep2	BMAA	pathogenic_control	2	batch_2
BMAA_rep4	BMAA	pathogenic_control	4	batch_2
BMAA_rep5	BMAA	pathogenic_control	5	batch_2
BMAA_rep6	BMAA	pathogenic_control	6	batch_2
BMAAals_rep1	CNR401	treated	1	batch_1
BMAAals_rep2	CNR401	treated	2	batch_1
BMAAals_rep3	CNR401	treated	3	batch_2
BMAAals_rep4	CNR401	treated	4	batch_2
BMAAals_rep5	CNR401	treated	5	batch_2
BMAAals_rep6	CNR401	treated	6	batch_2
BMAAeda_rep1	Edaravone	treated	1	batch_2
BMAAeda_rep2	Edaravone	treated	2	batch_2
BMAAeda_rep3	Edaravone	treated	3	batch_1
BMAAeda_rep4	Edaravone	treated	4	batch_1
BMAAeda_rep5	Edaravone	treated	5	batch_2
BMAAeda_rep6	Edaravone	treated	6	batch_2
BMAApure_rep1	CFA	treated	1	batch_2
BMAApure_rep2	CFA	treated	2	batch_1
BMAApure_rep3	CFA	treated	3	batch_2
BMAApure_rep4	CFA	treated	4	batch_2
BMAApure_rep6	CFA	treated	6	batch_2

Statistical Design

- A single `DeseqDataSet` was fit, so that there was one global GLM per gene for all 22 samples. This allows for more statistical power when estimating gene-wide dispersions since more replicates are available, helping with controlling the noise in this messy dataset.
- A reference-means model was used with BMAA and batch_1 as the reference levels (intercept; β_0). The three drugs were each given a coefficient referenced to this intercept ($\beta_2, \beta_3, \beta_4$) with the option to toggle on batch_2 (β_1), effectively adjusting for batch effects and isolating each treatment coefficient. This setup aligns with the Drosophila analysis where results were in the form of treatment tested against a pathological control as reference.

The design matrix used in this analysis.

sample	Intercept (β_0)	batch[T.batch_2] (β_1)	drug[T.CNR401] (β_2)	drug[T.Edaravone] (β_3)	drug[T.CFA] (β_4)
BMAA_rep1	1.0	0	0	0	0
BMAA_rep2	1.0	1	0	0	0
BMAA_rep4	1.0	1	0	0	0
BMAA_rep5	1.0	1	0	0	0
BMAA_rep6	1.0	1	0	0	0
BMAAals_rep1	1.0	0	1	0	0
BMAAals_rep2	1.0	0	1	0	0
BMAAals_rep3	1.0	1	1	0	0
BMAAals_rep4	1.0	1	1	0	0

BMAAals_rep5	1.0	1	1	0	0
BMAAals_rep6	1.0	1	1	0	0
BMAAeda_rep1	1.0	1	0	1	0
BMAAeda_rep2	1.0	1	0	1	0
BMAAeda_rep3	1.0	0	0	1	0
BMAAeda_rep4	1.0	0	0	1	0
BMAAeda_rep5	1.0	1	0	1	0
BMAAeda_rep6	1.0	1	0	1	0
BMAApure_rep1	1.0	1	0	0	1
BMAApure_rep2	1.0	0	0	0	1
BMAApure_rep3	1.0	1	0	0	1
BMAApure_rep4	1.0	1	0	0	1
BMAApure_rep6	1.0	1	0	0	1

The contrasts used in this analysis, of format ['variable_of_interest', 'tested_level', 'ref_level'].

["drug", "CNR401", "BMAA"]

["drug", "CFA", "BMAA"]

["drug", "Edaravone", "BMAA"]